① ~~Cyclic modules are acted on by scalar by $Z_{\mathfrak{g}}$.~~

Highest weight modules: need a generating vector that is also highest weight

(i.e. $P(-2)$ is not $Z_{\mathfrak{g}}$-semisimple i.e. indecomposible $\Rightarrow$ cyclic not highest weight)

$M(0)$

$M(-2)$

Exercise: Humphrey's BGG $\mathcal{O}$

$0 \to M(0) \to P(-2) \to M(-2) \to 0$

② $P(-2)$ D-module $\quad D/\cancel{\partial x \partial} \quad D/x\partial x$

decomposible

2nd order diff. equation

$\log(x)$, $x$ L.I solutions

$\partial x \partial(f) = 0 \quad f = A\log x + B$

$D/\partial x \partial \longrightarrow D/x\partial - 1 \oplus D/\partial$ ($A\log x$, $B$)

$1 \longmapsto (\partial x - 1,\ 1 - x\partial)$

$A + B \longleftarrow (A, B)$

$0 \to M(0) \to P(-2) \to M(-2) \to 0 \qquad 0 \to D/x\partial \to D/x\partial x \to D/x \to 0$ $\quad \triangle$ const.

$0 \to M^\vee(-2) \to P(-2) \to M^\vee(0) \to 0 \qquad 0 \to D/x \to D/x\partial x \to D/\partial x \to 0$ $\quad \nabla$ const.

$\int \delta$ "=" $H(x)$ Heaviside "step function"

$x\partial \qquad x\partial x \qquad \frac{1}{x}H(x)$

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Last time: an our way proof: $\mathcal{O}(\tilde{N}) = \mathcal{O}(N)$ $\quad H^i(\tilde{N}, \mathcal{O}_{\tilde{N}}) = 0 \quad i > 0$

Lemma: $f: X \to Y$ proper, birational, $Y$ aff. normal, then $f_*\mathcal{O}_X = \mathcal{O}_Y$

① $\tilde{N} = G \overset{B}{\times} \mathfrak{n}$ $\qquad \tilde{N}$ reduced $\rightsquigarrow$ $\mathcal{A}(\tilde{N})$ reduced

$\tilde{N} \to \mathfrak{g}$, $N \hookrightarrow \mathfrak{g}$

trivializing section coming from $G$ action on $\mathfrak{g}$

② $\mu: \tilde{N} \to N$ is proper: $\qquad \tilde{N} = G \overset{B}{\times} \mathfrak{n} \hookrightarrow G \overset{B}{\times} \mathfrak{g} \simeq G/B \times \mathfrak{g} \twoheadrightarrow \mathfrak{g}$

③ $N$ is normal: ① $N$ is complete intersection $\Rightarrow$ Cohen-Macaulay.

$N \to \mathfrak{g}$

$\downarrow \quad \downarrow$

$\{0\} \to \mathfrak{h}/W$ $\longleftarrow$ projection

expected dim: $\dim(G) - \dim(T) = K$

$\dim(N) \geq K$

every nilpotent is in a Borel $\Longrightarrow$ $\mu$ surjective, $\dim(N) \leq K$.

$\mu: \tilde{N} \to N$

$\dim(\tilde{N}) = \dim(G/B) + \dim(\mathfrak{n})$

$= \dim(G) - \dim(B) + \dim(\mathfrak{n}) = \dim(G) - \dim(T) = K$

So $\dim(N) = K$.

② Serre's criterion: ✓ • $X$ CM

• regular in codim 2 $\Rightarrow$ $X$ normal

singular locus in codim 2

Assume: $G$ finitely many orbits in $\mathcal{N}$

For $GL_n$: Jordan decomp. of nilpotent matrices $\updownarrow$ partitions of $n$

Claim: orbits are even $\Rightarrow$ stratify by orbits (codim 2)

$\mathcal{N}$ irreducible b/c $\tilde{\mathcal{N}}$ is

Kirillov: put a symplectic form on each orbit

$\mathcal{O} \subseteq \mathcal{N} \subseteq \mathfrak{g}^*$ $x \in \mathfrak{g}^*$

$\omega_x(X \cdot x, Y \cdot x) = \langle x, [X,Y] \rangle$

Exercise: show well-def, nondeg, closed

$\Rightarrow \mathcal{N}$ is normal.

· Birational: $GL_n$: $\begin{pmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & 0 & 1 & \\ & & & 0 & 1 \\ & & & & 0 \end{pmatrix}$ regular nilpotent

fixes exactly one Borel $B$

$e_{n-1} \to e_{n-1} \to \cdots \to e_1 \to 0$ in flag: $\langle e_1 \rangle \subseteq \langle e_1, e_2 \rangle \subseteq \cdots$

$Z = \tilde{\mathcal{N}} \times_{\mathcal{N}} \tilde{\mathcal{N}}$

$H_\bullet^{Borel}(Z)$ Springer convolution alg.

$H_{top}^{Borel}(Z) \simeq \mathbb{C}[W] \curvearrowright H_{top}(\tilde{\mathcal{N}})$

see Irreps by fibers

$G \curvearrowright X$

Def A $G$-equivariant sheaf on $X$ is a sheaf $\mathcal{F}$ w/ an equivariance str. map $\phi: a^*\mathcal{F} \xrightarrow{\simeq} p^*\mathcal{F}$ satisfying a cocycle condition.

$$G \times G \times X \overset{p}{\underset{a}{\overset{m}{\rightrightarrows}}} G \times X \overset{p}{\underset{a}{\rightrightarrows}} X \qquad \mathcal{F}$$

$$\begin{tikzcd} x & x \arrow[r] & gx \arrow[l, "\phi"'] \\ & (g,x) & (g_1, g_2x) \end{tikzcd}$$

$$\begin{tikzcd} & g_2x \arrow[dr, "a"] & \\ x \arrow[ur, "p"] \arrow[rr, "m"'] & & g_1g_2x \\ & (g_1, g_2x) & \end{tikzcd}$$

$$\boxed{p^*\phi \circ a^*\phi = m^*\phi}$$

$$\begin{tikzcd} (a^*\mathcal{F})_{(g,x)} \arrow[d, "\phi"'] & = & \mathcal{F}_{g \cdot x} \arrow[d] \\ (p^*\mathcal{F})_{(g,x)} & = & \mathcal{F}_x \end{tikzcd}$$

$f: \mathcal{F} \to \mathcal{G}$ is equiv. if $a^*\phi = p^*\phi$

**Def.** A weakly equiv. D-module on $X$ is a quasicoh. $G$-eq sheaf $\mathcal{M}$ on $X$ + $D_X$-module structure on $\mathcal{M}$ such that structure map $D_X \otimes \mathcal{M} \to \mathcal{M}$ is $G$-eq mor.

$x \in \mathfrak{g}$ $\quad$ ($\mathcal{M}$ eq sheaf) $\quad \alpha_x : \mathcal{M} \to \mathcal{M}$

$$\alpha^* \mathcal{M} \xrightarrow[\theta]{\phi} p^* \mathcal{M} \simeq \mathcal{O}_G \boxtimes \mathcal{M} \xrightarrow[k]{x \cdot} \mathcal{O}_G \boxtimes \mathcal{M}$$

restrict to $\{e\} \times X$: $\quad \mathcal{M} \longrightarrow \mathcal{M}$ $\quad$ $k$-mor.

$\mathcal{M}$ D-module $\quad \delta_x : \mathcal{M} \to \mathcal{M}$

$U\mathfrak{g} \longrightarrow D_X$

$x \longmapsto \delta_x$

$a : G \times X \to X$

$\mathfrak{g} \otimes \mathcal{O}_G \boxtimes T_X \longrightarrow T_X$

$((x \otimes 1), 0) \longmapsto$

**Def.** $\mathcal{M}$ is string eq if weak eq + $\alpha_x = \delta_x$ $\forall x \in \mathfrak{g}$

**Ex.** weak eq. $\quad p : i^* \mathcal{M} \to j^* \mathcal{M}$ is $\mathcal{O}_G \boxtimes D_X$-linear.

string. $\quad D_G \boxtimes D_X$-linear.

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**Ex.** $\mathbb{G}_m$-eq. D-mod on $*$.

$V = \bigoplus V_n$

weak: graded vector space.

$z \cdot v_n = z^n v_n$

string: $\delta_x = 0$

Compute: $\alpha_x$ on $V_n$ by $n$

$\frac{d}{dz}(z^n v_n) = n z^{n-1} v_n \Big|_{z=1}$

$= n \cdot v_n$

$$v \otimes z \longmapsto v \otimes z^n \longmapsto v \otimes n z^{n-1}$$

$\xi z = 1$

$v \longmapsto nv$

$V = \bigoplus V_n$ if $\boxed{n = 0}$

$\boxed{V = V_0}$ i.e. $V$ is a vector space.